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Project Report

on

Graph Pathfinding Visualizer

Submitted by

ANMOL

25MCI10382

Under the Guidence Of

Dr. Amanpreet Kaur Sandhu

partial fulfilment for the award of the degree of

MASTER'S OF COMPUTER APPLICATION IN

ARTIFICIAL INTELLIGANCE & MACHINE LEARNING

CHANDIGARH UNIVERSITY

November 2025

Certificate

This is to certify that **Anmol Rana**, bearing UID: **25MCI10382**, of University Institute of Computing, has successfully completed the mini project entitled **"Student Performance Dashboard using Machine Learning"**

under the guidance of **Dr Amanpreet Kaur Sandhu** during the academic year **2025–2026** in partial fulfillment of the requirements for the course **Artificial Intelligence & Machine Learning**.

This project demonstrates the implementation of Machine Learning techniques to analyze and predict student performance using various parameters such as study hours, attendance, and academic records. The project also includes an interactive dashboard developed using Python libraries for data visualization and insights generation.

The student has shown dedication, creativity, and a clear understanding of Machine Learning concepts, data analysis, and visualization techniques throughout the development of this project.

We hereby declare that this work is genuine and has been carried out by the student as part of the prescribed curriculum.

Project Guide:

Dr.Amanpreet Kaur Sandhu

Acknowledgement

I would like to express my sincere gratitude to my respected faculty member, **Dr Amanpreet Kaur Sandhu** for their valuable guidance, constant encouragement, and continuous support throughout the development of this mini project titled **"Student Performance Dashboard using Machine Learning."** Their expert advice, insightful feedback, and motivation helped me to explore the subject in depth and successfully complete this project.

I am also extremely thankful to my institution, **University Institute of Computing and Chandigarh University**, for providing the necessary facilities, learning resources, and an excellent academic environment that enabled me to carry out this work effectively.

This project gave me an opportunity to enhance my knowledge of Machine Learning, data analysis, and data visualization. I gained practical experience in working with real datasets, implementing algorithms such as Linear Regression, and creating interactive dashboards using Python libraries like Pandas, Streamlit, and Plotly.

I would also like to extend my heartfelt thanks to my classmates, friends, and family members for their cooperation, constant support, and encouragement throughout the project development process. Their motivation played a significant role in the successful completion of this work.

Finally, I express my sincere gratitude to everyone who directly or indirectly contributed to the successful completion of this mini project.

Anmol

MCA [AI&ML]

UID:- 25MCI10382

Chandigarh University

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Abstract

The **Student Performance Dashboard using Machine Learning** is an interactive and data-driven project designed to analyze and predict student academic performance using various influencing factors. In today's educational environment, understanding student performance is essential for improving learning outcomes and identifying areas where students need support. This project aims to bridge the gap between raw academic data and meaningful insights through visualization and predictive modeling.

The project is implemented using Python and incorporates key libraries such as Pandas for data manipulation, Plotly for data visualization, and Streamlit for building an interactive web-based dashboard. The dataset includes attributes such as study hours, attendance, previous scores, and other academic indicators, which are used to train a Machine Learning model.

A Linear Regression algorithm is applied to predict student performance based on the given input features. The dashboard allows users to explore the dataset through graphs, charts, and visual analytics, making it easier to identify patterns and trends in student behavior and outcomes. Users can also input custom values to obtain predicted results, enhancing the interactivity and usefulness of the application.

Through this project, users gain a better understanding of how Machine Learning can be applied in the education sector to support decision-making and performance evaluation. It combines the concepts of data analysis, predictive modeling, and visualization into a single integrated system.

In conclusion, the Student Performance Dashboard serves as an effective tool for analyzing academic performance and demonstrates the practical application of Machine Learning techniques in solving real-world problems in education.

Introduction

In the modern era of education, data plays a crucial role in understanding and improving student performance. Educational institutions generate a vast amount of academic data, but extracting meaningful insights from this data can be challenging without the use of advanced analytical tools. Machine Learning provides powerful techniques to analyze such data and make accurate predictions, helping educators and students make informed decisions.

The project titled **"Student Performance Dashboard using Machine Learning"** focuses on analyzing and predicting student performance based on various academic and behavioral factors. These factors may include study hours, attendance, previous academic scores, and other relevant parameters that influence a student's overall performance. By utilizing Machine Learning algorithms, the project aims to identify patterns and relationships within the data that are not easily visible through traditional methods.

This project uses **Linear Regression**, a fundamental supervised learning algorithm, to predict student performance. The implementation is done using Python along with libraries such as Pandas for data handling, Plotly for visualization, and Streamlit for creating an interactive and user-friendly dashboard. The dashboard enables users to explore the dataset visually through charts and graphs, as well as input custom values to generate performance predictions.

The main objective of this project is to demonstrate how Machine Learning can be effectively used in the field of education to enhance performance analysis and support data-driven decision-making. It not only helps in understanding student behavior but also assists in identifying students who may require additional support or improvement.

Overall, this project integrates concepts of data analysis, predictive modeling, and visualization to create a comprehensive system that simplifies the process of analyzing student performance and provides valuable insights for academic improvement.

Tools and Technologies Used

❖ **Programming Language:**

- Python – Used for implementing Machine Learning algorithms, data processing, and backend logic of the application.

❖ **Machine Learning Algorithm:**

- Linear Regression – Used to predict student performance based on input features such as study hours, attendance, and previous scores.

❖ **Python Libraries:**

- Pandas – Used for data manipulation and analysis.
- NumPy – Used for numerical computations and handling arrays.
- Scikit-learn – Used for implementing the Linear Regression model and training the dataset.
- Plotly – Used for creating interactive graphs and visualizations.
- Streamlit – Used to build an interactive web-based dashboard.

❖ **Development Environment:**

- Jupyter Notebook / Visual Studio Code – Used for writing, testing, and debugging the code.

❖ **Dataset:**

- CSV File (Student Performance Dataset) – Contains student-related data such as study hours, attendance, and marks used for training and testing the model.

Delimiter: ,

	student_id	weekly_self_study_hours	attendance_percentage	class_participation	total_score	grade
1	1	18.5	95.6	3.8	97.9	A
2	2	14.0	80.0	2.5	83.9	B
3	3	19.5	86.3	5.3	100.0	A
4	4	25.7	70.2	7.0	100.0	A
5	5	13.4	81.9	6.9	92.0	A
6	6	13.4	65.1	5.0	97.5	A
7	7	26.1	81.8	5.9	100.0	A
8	8	20.4	100.0	4.0	96.1	A
9	9	11.7	100.0	8.2	69.8	C
10	10	18.8	67.6	6.0	80.3	B
11	11	11.8	82.3	4.6	66.4	C
12	12	11.7	75.9	7.6	72.5	B
13	13	16.7	77.9	2.5	95.2	A
14	14	1.6	92.1	5.7	60.9	C
15	15	2.9	90.7	5.4	52.1	D
16	16	11.1	87.4	3.9	73.5	B
17	17	7.9	85.9	7.7	63.0	C
18	18	17.2	74.0	8.1	100.0	A

❖ Visualization Tools:

- Interactive charts and graphs (bar charts, line graphs, scatter plots) are used to represent student performance trends and insights effectively.

❖ Operating System:

- Compatible with Windows, Linux, and macOS systems.

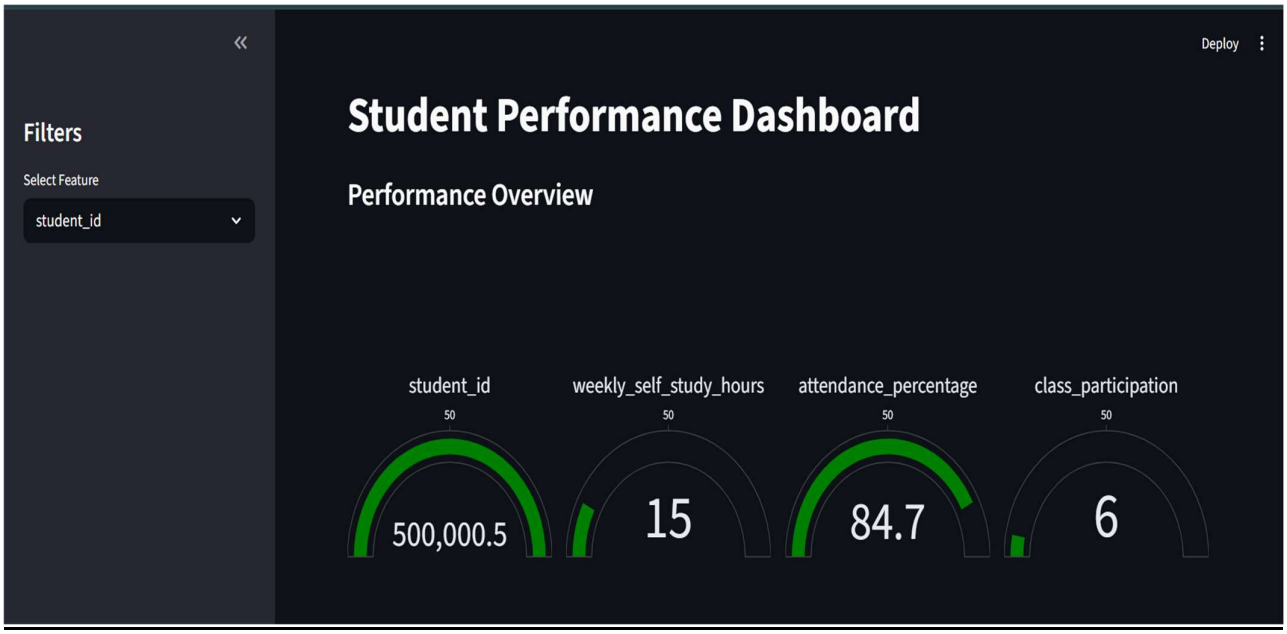
Source Code

```
import pandas as pd
import streamlit as st
import plotly.express as px
import plotly.graph_objects as go
from sklearn.linear_model import LinearRegression

# ---------------- CONFIG ----------------
st.set_page_config(page_title="Student Dashboard", layout="wide")

# ---------------- LOAD DATA ----------------
df = pd.read_csv("student_performance.csv")
df.columns = df.columns.str.lower()

numeric_df = df.select_dtypes(include=['int64','float64'])

# ---------------- MODEL ----------------
X = numeric_df.iloc[:, :-1]
y = numeric_df.iloc[:, -1]

model = LinearRegression()
model.fit(X, y)

# ---------------- SIDEBAR ----------------
st.sidebar.title(" Filters")

selected_col = st.sidebar.selectbox("Select Feature", numeric_df.columns)

# ---------------- MAIN TITLE ----------------
st.title(" Student Performance Dashboard")

# ---------------- GAUGE FUNCTION ----------------
def gauge(title, value):
    return go.Figure(go.Indicator(
```

```
        mode="gauge+number",
        value=value,
        title={'text': title},
        gauge={'axis': {'range': [0, 100]}}
    ))

# ---------------- TOP METRICS ----------------
st.subheader(" Performance Overview")

cols = st.columns(4)
means = numeric_df.mean()

for i in range(min(4, len(means))):
    cols[i].plotly_chart(gauge(means.index[i], means[i]), width='stretch')

# ---------------- CHART ----------------
st.subheader(" Feature Analysis")

fig = px.histogram(df, x=selected_col)
st.plotly_chart(fig, width='stretch')

# ---------------- BAR ----------------
st.subheader(" Average Scores")

fig_bar = px.bar(x=means.index, y=means.values, color=means.values)
st.plotly_chart(fig_bar, width='stretch')

# ---------------- PREDICTION ----------------
st.subheader(" Predict Student Score")

col1, col2 = st.columns(2)

inputs = []

for i, col in enumerate(X.columns):
    if i % 2 == 0:
```

```
        val = col1.number_input(f"{col}", value=0)
    else:
        val = col2.number_input(f"{col}", value=0)
    inputs.append(val)

if st.button(" Predict"):
    pred = model.predict([inputs])
    st.success(f" Predicted Score: {round(pred[0],2)}")

# ---------------- DATA ----------------
st.subheader(" Dataset Preview")
st.dataframe(df.head())
```

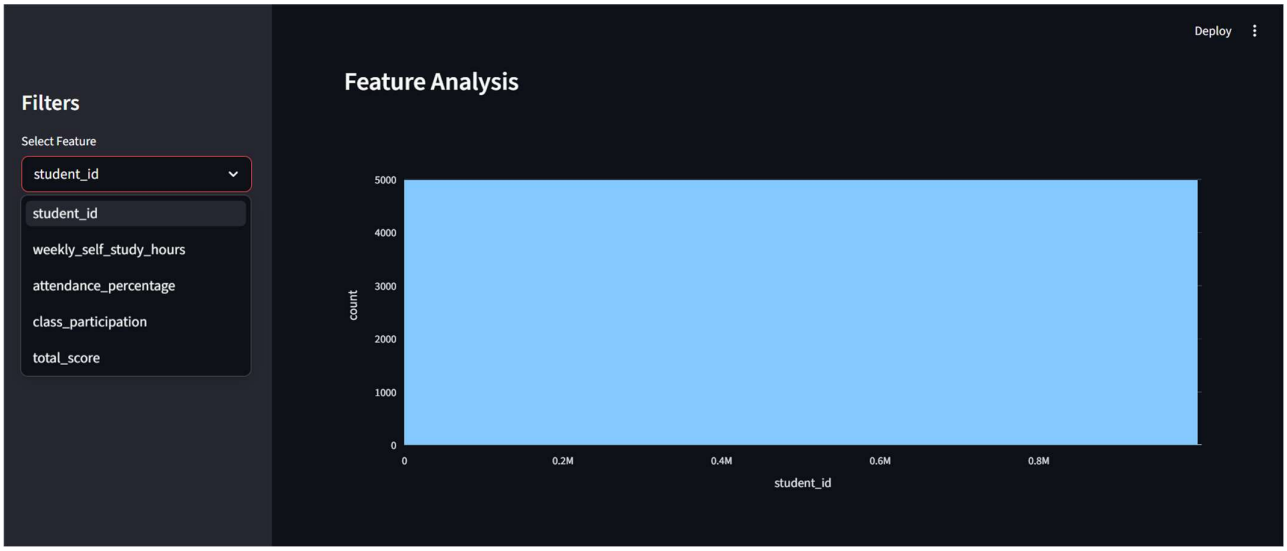
Result



Dashboard Overview (Fig. 1)

This view shows the main dashboard of the Student Performance System.

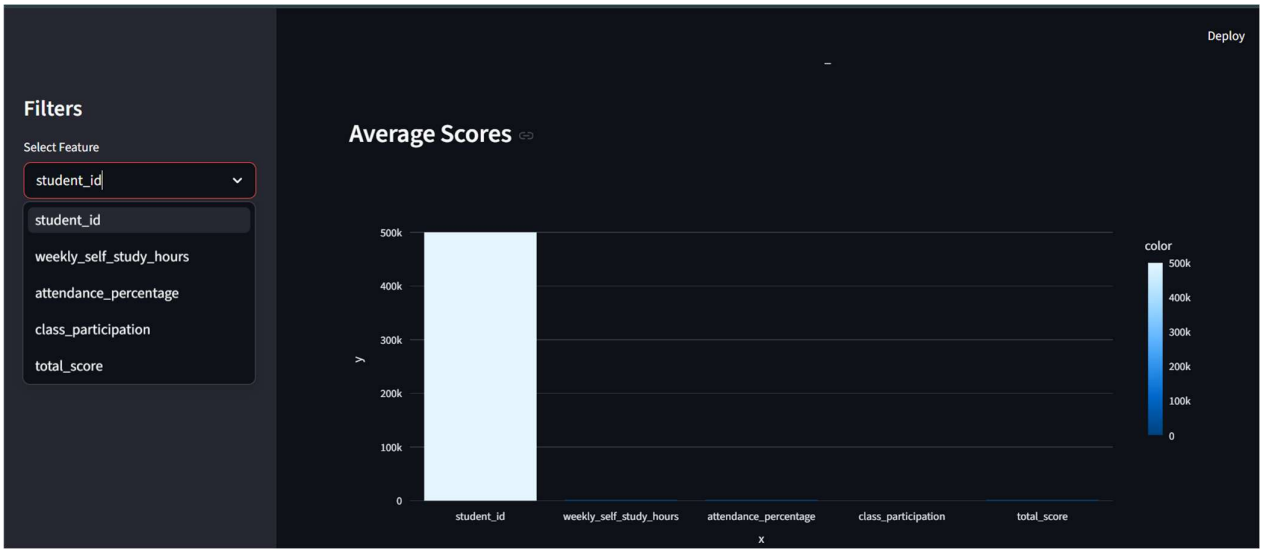
- Displays key performance indicators using gauge charts.
- Shows values like study hours, attendance, and participation.
- Provides a quick overview of student performance.



Feature Analysis View (Fig. 2)

This view represents data analysis using graphs.

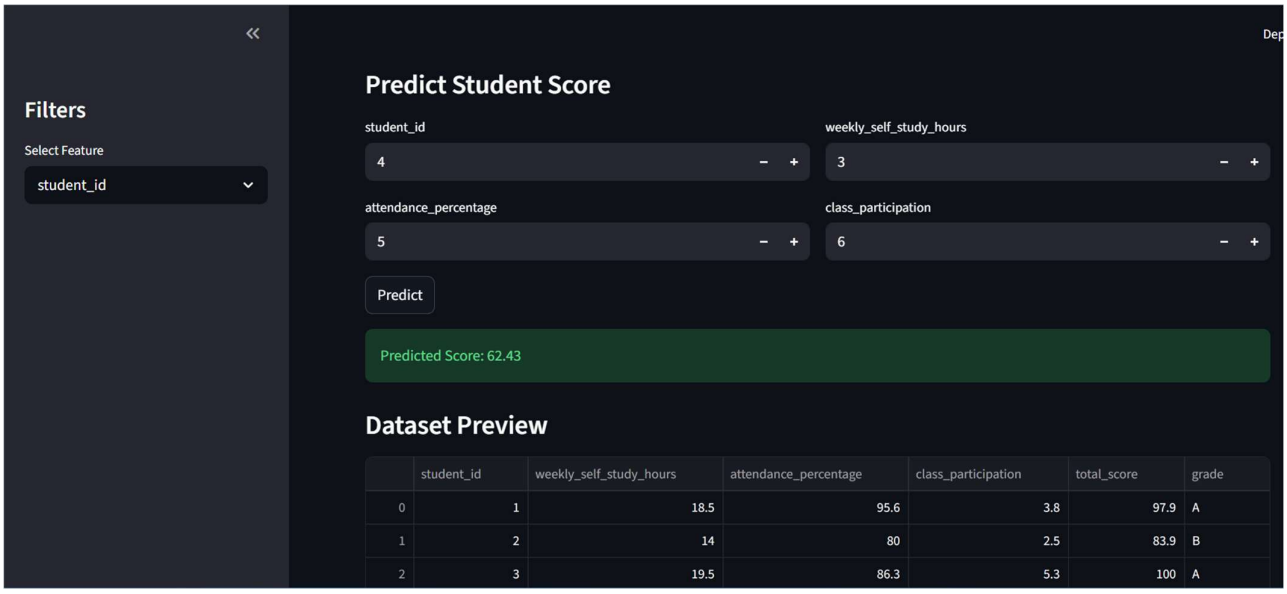
- Users can select features from the filter panel.
- Graphs like bar charts help in understanding data distribution.



Average Scores View (Fig. 3)

This view shows comparison of different features.

- Displays average values using bar charts.
- Helps in comparing performance factors.
- Makes it easy to understand overall trends.



Prediction & Dataset View (Fig. 4)

This view focuses on prediction and data preview.

- Users input values to predict student score.
- Displays predicted result using Machine Learning model.
- Shows dataset table for reference and comparison.

Implementation Steps

Step 1: Problem Definition

The objective of this project was to develop a system that can analyze student data and predict their performance using Machine Learning techniques.

Step 2: Data Collection

A dataset was collected in CSV format containing student details such as study hours, attendance, class participation, and total scores.

Step 3: Data Preprocessing

The dataset was cleaned by removing missing or unnecessary values. Relevant features were selected for analysis and prediction.

Step 4: Model Selection and Training

A Linear Regression algorithm was used to train the model. The model learns the relationship between input features and student performance.

Step 5: Dashboard Development

An interactive dashboard was created using Streamlit to display data and allow user interaction.

Step 6: Data Visualization

Graphs such as bar charts and analysis plots were created using Plotly to visualize trends and patterns in student performance.

Step 7: Prediction System Integration

Input fields were added where users can enter values like study hours, attendance, and participation. The trained model predicts the student score based on these inputs.

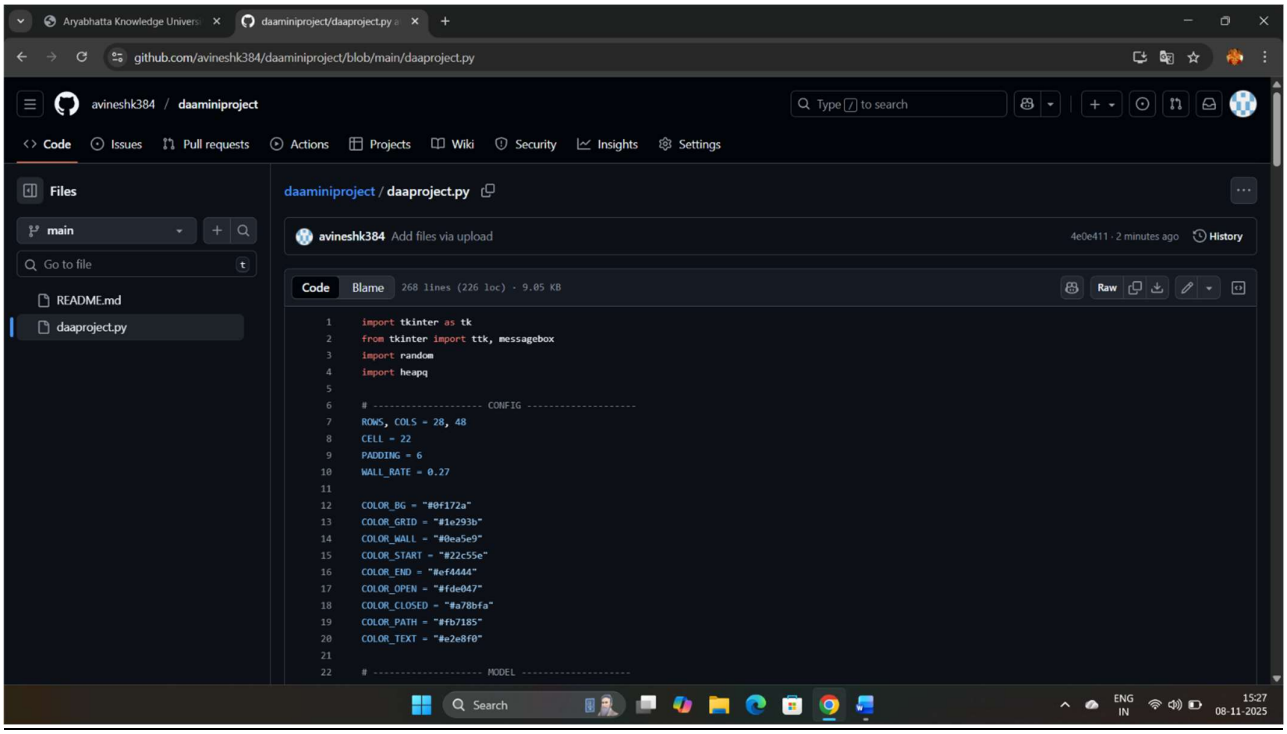
Step 8: Testing and Validation

The system was tested with different inputs to ensure accuracy and proper functioning of all features.

Step 9: Final Output

The final system provides an interactive dashboard that combines data analysis, visualization, and prediction of student performance.

GitHub Screenshot



Link:- https://github.com/avineshk384/daaminiproject

Conclusion

The Student Performance Dashboard using Machine Learning project successfully demonstrates how data analysis and predictive modeling can be used to evaluate student performance. The system effectively analyzes student data such as study hours, attendance, and class participation to generate meaningful insights.

By using Linear Regression, the project is able to predict student scores based on input features. The integration of an interactive dashboard using Streamlit and visualizations using Plotly makes the system user-friendly and easy to understand.

This project helps in identifying patterns and relationships between different academic factors, which can support better decision-making in the education system. It also shows how Machine Learning can be applied to real-world problems in a simple and effective way.

Overall, the project achieves its objective of combining data visualization and prediction into a single platform, making it a useful tool for analyzing and improving student performance.

References

- Python Official Documentation – https://docs.python.org/3/
- Scikit-learn Documentation – https://scikit-learn.org/
- Streamlit Documentation – https://docs.streamlit.io/
- Plotly Documentation – https://plotly.com/python/
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